

Fengping River Hydropower — Main Analysis

Financial Feasibility Frontier under Hydrological Variability

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1 Sediment and Rating Curve

1.1 Suspended Sediment Rating Curve

The suspended sediment load (SSL, MT/day) is estimated from daily discharge Q (cms) via a power-law rating curve calibrated on 492 field observations (1959–2023):

$$SSL = a \cdot Q^b$$

fitted by ordinary least squares in log-log space. Only observations flagged ok with detectable sediment concentration are used; below-detection-limit records are excluded.

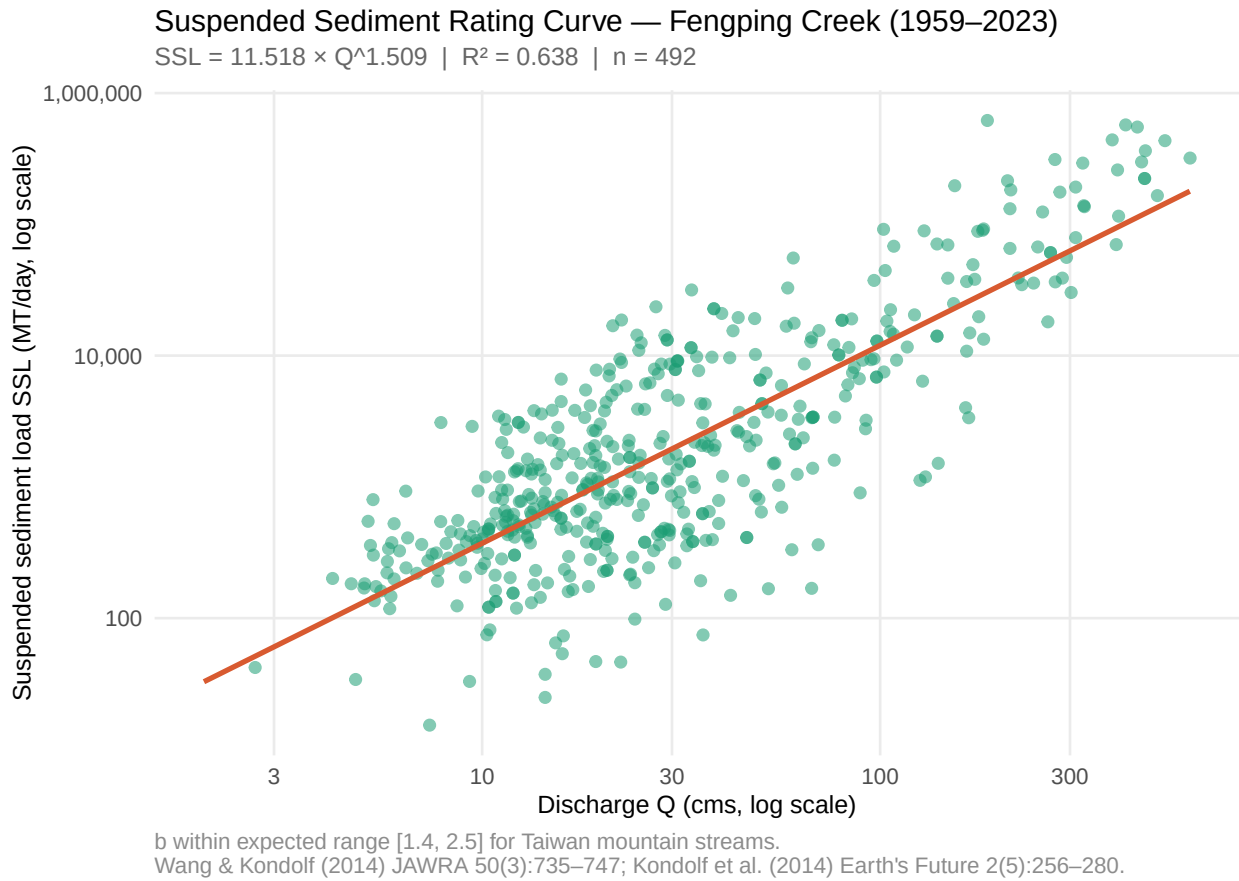


Figure 1: Suspended sediment rating curve for Fengping Creek. Each point is one field sampling event ($n = 492$, flag == 'ok'). The exponent $b = 1.51$ falls within the range [1.4, 2.5] documented for Taiwan mountain streams (Wang & Kondolf 2014; Kondolf et al. 2014).

1.2 Reservoir Sedimentation — W1 and W2

Trap efficiency TE is computed annually using the Brune (1953) curve:

$$TE_t = \frac{CI_t}{CI_t + 0.0021}, \quad CI_t = \frac{S_t}{V_{\text{annual}}}$$

where S_t is the remaining effective storage (m^3) and V_{annual} is the mean annual inflow volume.

Weir Sedimentation Over 35-Year Project Lifetime



Brune (1953) trap efficiency curve approximation.
Deposit bulk density = 1300 kg/m^3 (Wang et al. 2018, Water 10(8):1034)

Figure 2: Simulated trap efficiency (TE) and remaining effective storage for W1 and W2 over the 35-year project lifetime. Brune (1953) curve approximation; deposit bulk density = 1300 kg/m^3 (Wang et al. 2018).

Table 1: Trap efficiency and remaining effective storage at key project milestones (years 1, 10, 20, 35).

Weir	Year	S remaining (m^3)	TE	V loss (m^3/yr)
W1 Lower	1	0	0.375	5,765,603
W1 Lower	10	0	0.000	0
W1 Lower	20	0	0.000	0
W1 Lower	35	0	0.000	0
W2 Upper	1	0	0.363	5,567,670
W2 Upper	10	0	0.000	0
W2 Upper	20	0	0.000	0
W2 Upper	35	0	0.000	0

2 Reservoir Simulation — Historical Baseline

The pondage simulation (`simulate_one_weir()`) uses a daily Euler forward water balance with a strict dispatch priority: (1) ecological flow, (2) power generation up to Q_{design} , (3) spillage.

The 24-hour forecast model (`simulate_one_weir_forward24()`) additionally looks ahead one day to decide how much water to retain, reducing spill during high-flow events while smoothing generation during droughts.

Both models are run on the full historical record (1958–2025) with the **recommended ecological flow** ($W1 = 5.0$ cms, $W2 = 1.0$ cms).

2.1 Operation Window — Interactive Date Selection

Change `START_DATE` and `END_DATE` to inspect any period of interest.

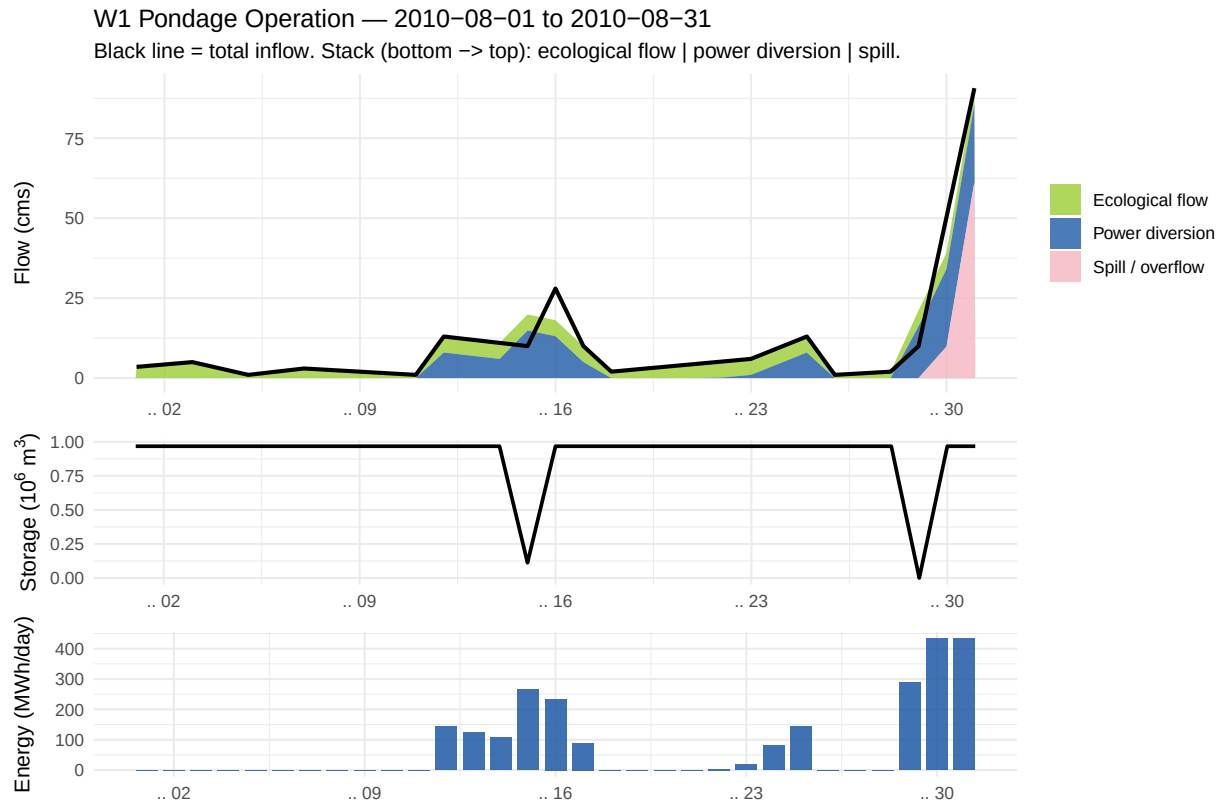


Figure 3: W1 reservoir operation during a representative typhoon-season window. Line = inflow; stacked areas = ecological flow, power diversion, and spill. Lower panels show storage trajectory and daily energy.

2.2 Annual Energy — Calendar Heatmap

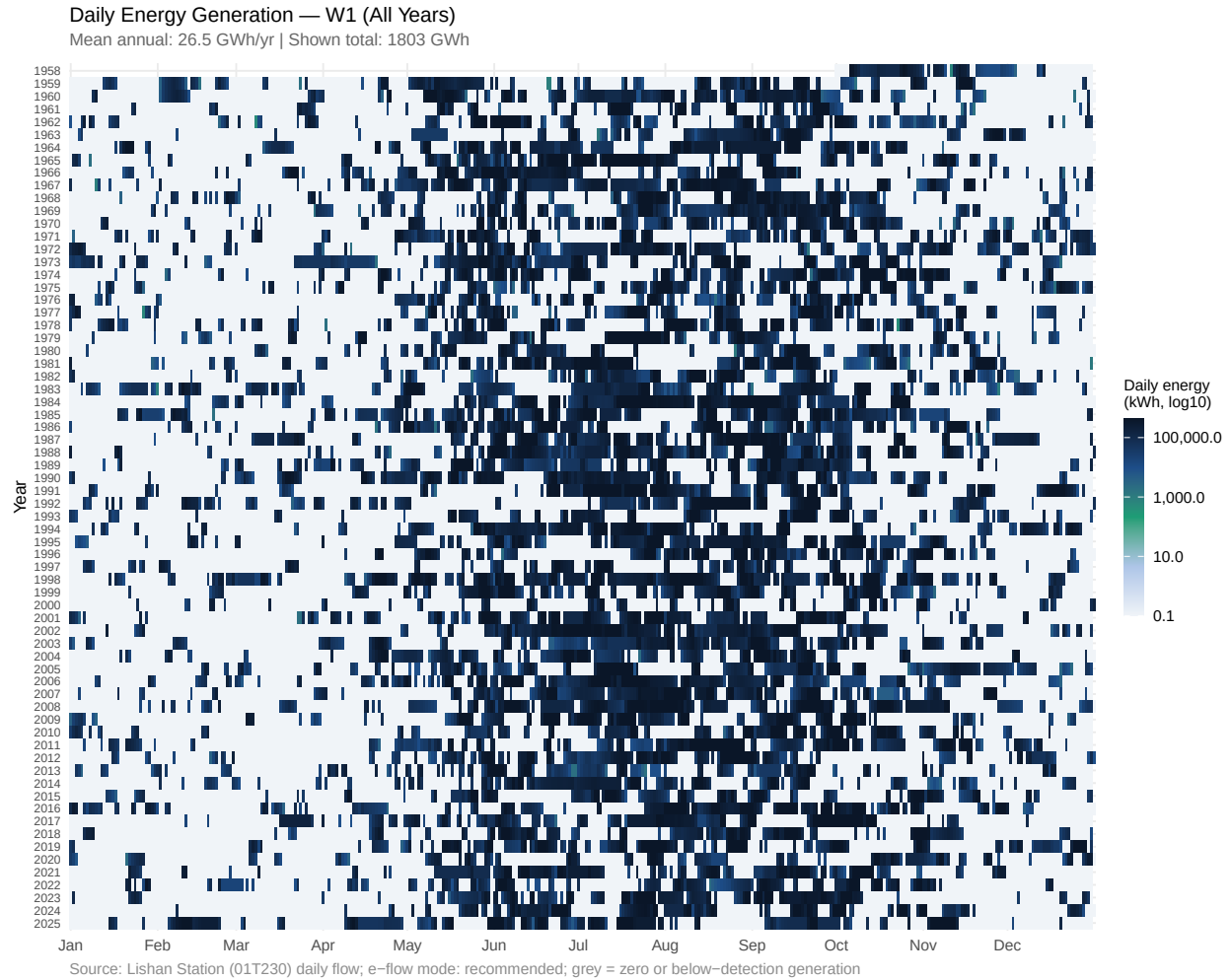


Figure 4: Calendar heatmap of daily energy generation for W1 (pondage mode, 1958–2025). Dark = high generation; grey = near-zero. Typhoon-season pulses visible as dark horizontal bands in July–September.

2.3 Annual Generation Summary — Run-of-River vs Pondage

Table 2: Annual generation statistics for W1 + W2 combined. Revenue computed at PPA rate (NTD 6.0/kWh). Run-of-river = immediate dispatch; Pondage = 24-hour forecast-informed. Historical record 1958–2025.

Mode	Mean			Mean Revenue (NTD B/yr)
	GWh/yr	P10 GWh/yr	P90 GWh/yr	
Run-of-river	45.2	29	58	0.271
Pondage (24-hr)	48.2	31	62	0.289

3 35-Year NPV Trajectory — Historical Baseline

NPV is computed using equity discount rate $r_e = 8\%$, LTV = 80%, and loan rate $r_\ell = 3\%$. Electricity revenue uses the PPA rate (NTD 6/kWh). Three CAPEX scenarios are compared:

- CAPEX_{OG} = NTD 5.0 billion (earliest public announcement)
- $\text{CAPEX}_{\text{EIA}}$ = NTD 6.4 billion (1999 EIA feasibility report)
- CAPEX_{2025} = NTD 9.7 billion (2025 investor conference)

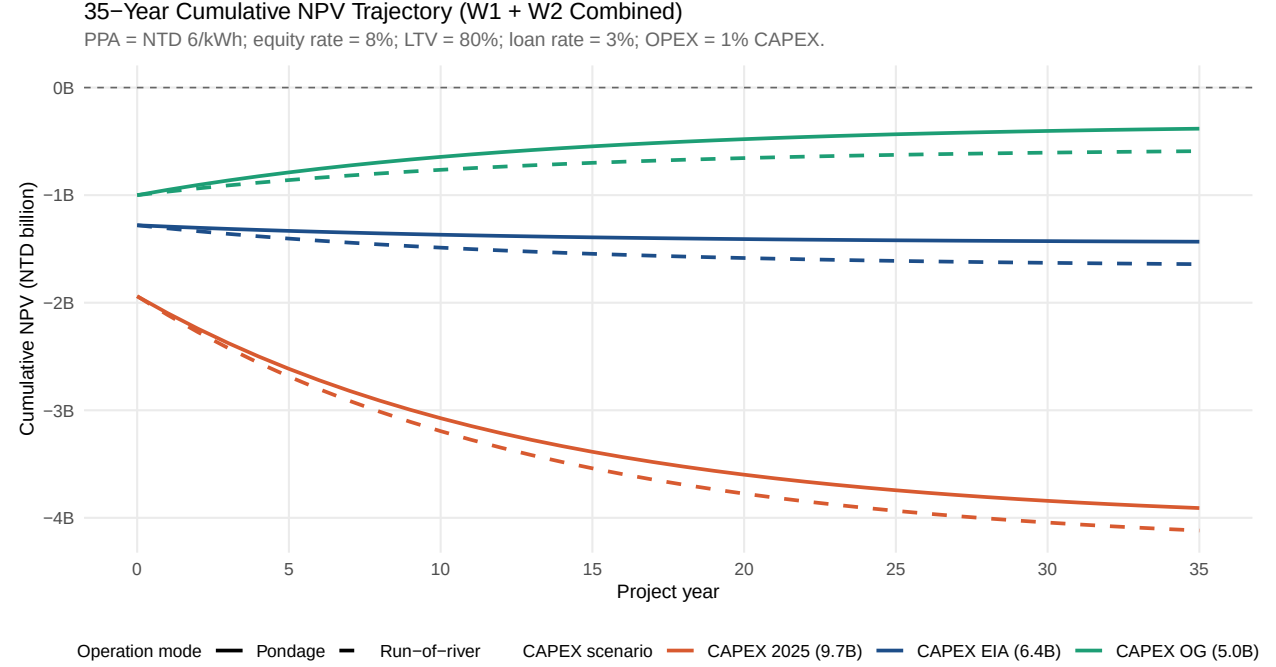


Figure 5: 35-year cumulative NPV trajectory for both operation modes and three CAPEX scenarios. Revenue at PPA rate (NTD 6.0/kWh). Dashed horizontal line = NPV breakeven.

4 Climate Bootstrap — Hydrological Variability Ensemble

4.1 Bootstrap Design

Two-layer bootstrap (Section 3 of `hydro_climate.R`):

1. **Layer 1 — Whole-year block bootstrap**: draw complete calendar years with replacement from the observed record (1958–2025), preserving within-year seasonal structure and typhoon event integrity.
2. **Layer 2 — CV scaling**: rescale daily flows around the annual mean to a target within-year CV level:

$$Q_{\text{scaled}}(t) = \mu_{yr} + (Q_{\text{boot}}(t) - \mu_{yr}) \times \lambda, \quad \lambda = \frac{CV_{\text{target}}}{CV_{\text{boot}}}$$

This preserves μ_{yr} exactly while enabling CV values beyond the historical range, consistent with SSP2-4.5 and SSP5-8.5 projections for eastern Taiwan (Shiau & Huang 2014; Tung et al. 2016; IPCC AR6 Ch. 11).

4.2 CV Scaling Validation

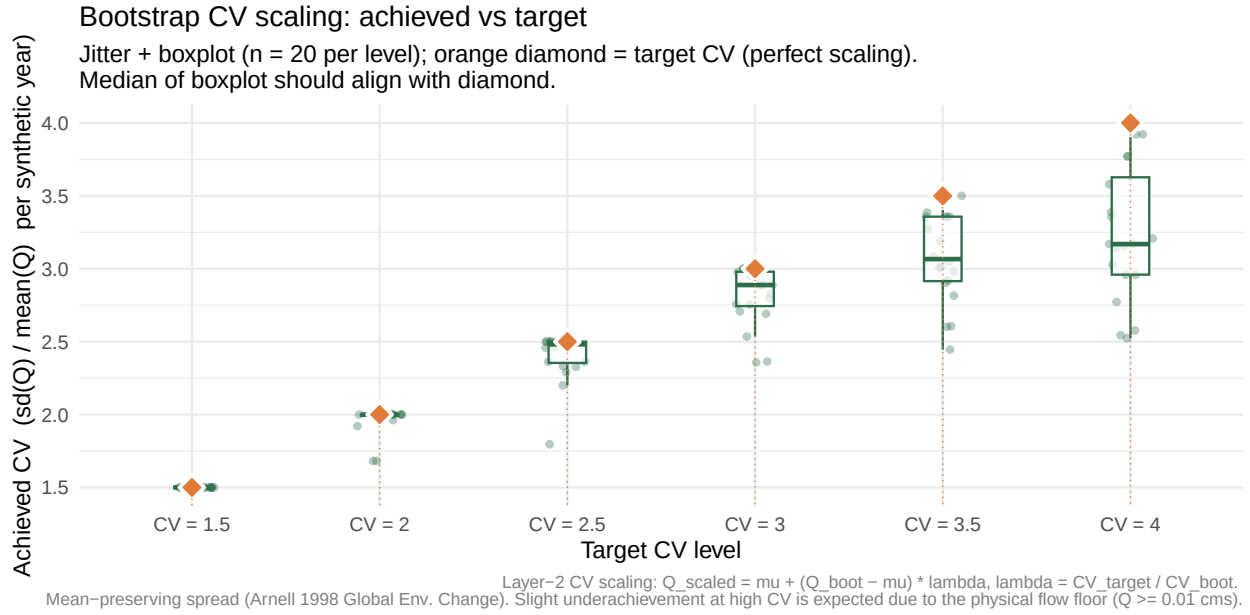


Figure 6: Validation of Layer-2 CV scaling. Violin plots show the distribution of achieved CV across 200 bootstrap replicates for each target CV level. Diamonds mark the target. The median should lie close to each target; slight underestimation at high CV levels is expected due to the physical flow floor constraint.

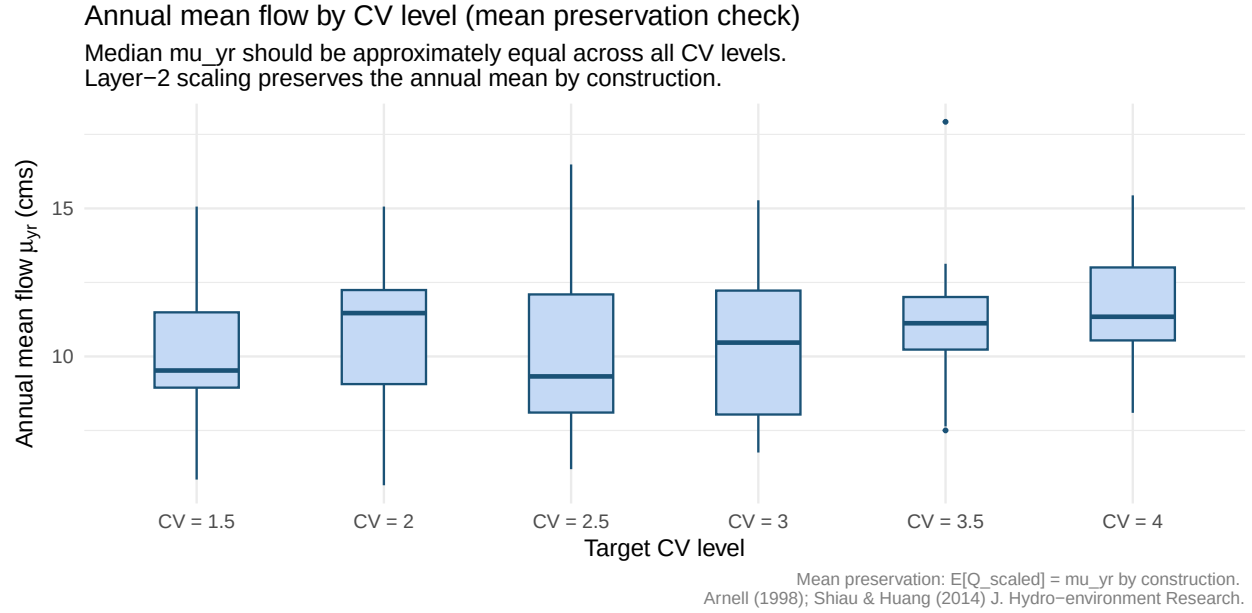


Figure 7: Mean-preservation check: annual mean flow (μ_{yr}) should be approximately constant across CV levels, confirming that Layer 2 does not alter the mean — only the variability.

4.3 Historical CV Trend

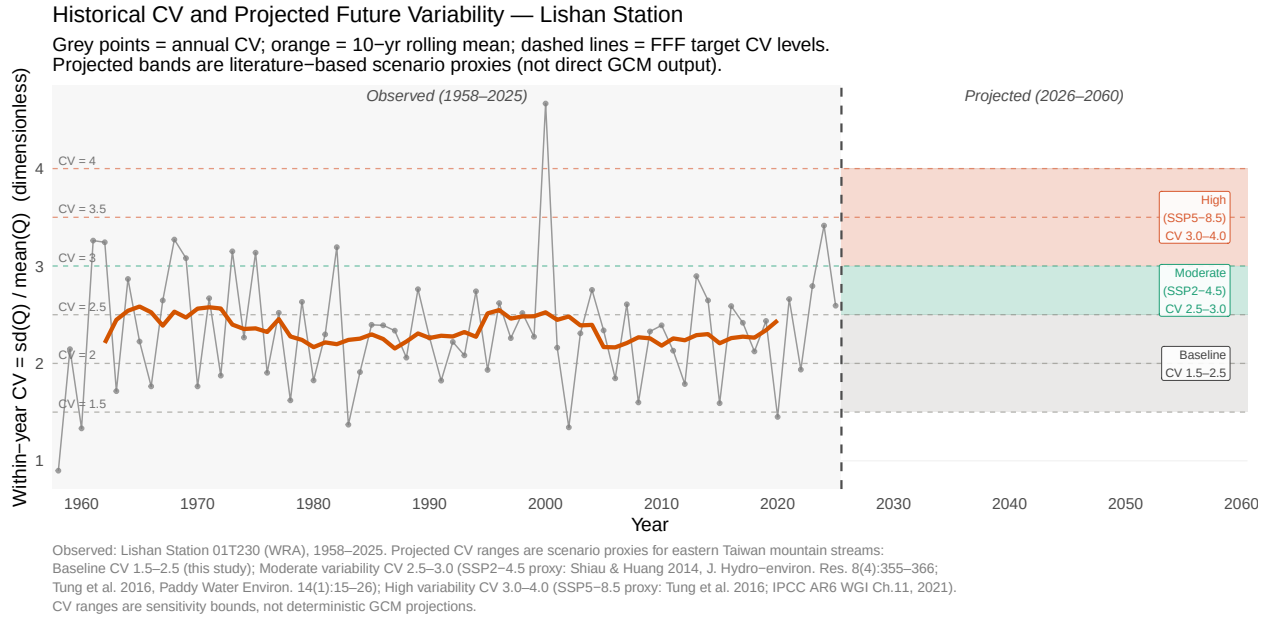


Figure 8: Historical within-year CV (1958–2025) and projected future variability scenario bands (2026–2060). Left zone: observed annual CV (grey points) with 10-year rolling mean (orange). Right zone: literature-based scenario proxies — Baseline (CV 1.5–2.5), Moderate/SSP2-4.5 (CV 2.5–3.0), High/SSP5-8.5 (CV 3.0–4.0). Vertical dashed line separates observed history from projected window. Sources: Shiau & Huang (2014); Tung et al. (2016); IPCC AR6 WGI Ch.11 (2021).

4.4 Bootstrap CV Distribution — Option A (target CV on X-axis)

Option A: Bootstrap CV distribution by target CV level

Points = individual bootstrap years; diamond = target (1:1 line).
Colour intensity = scenario variability level.

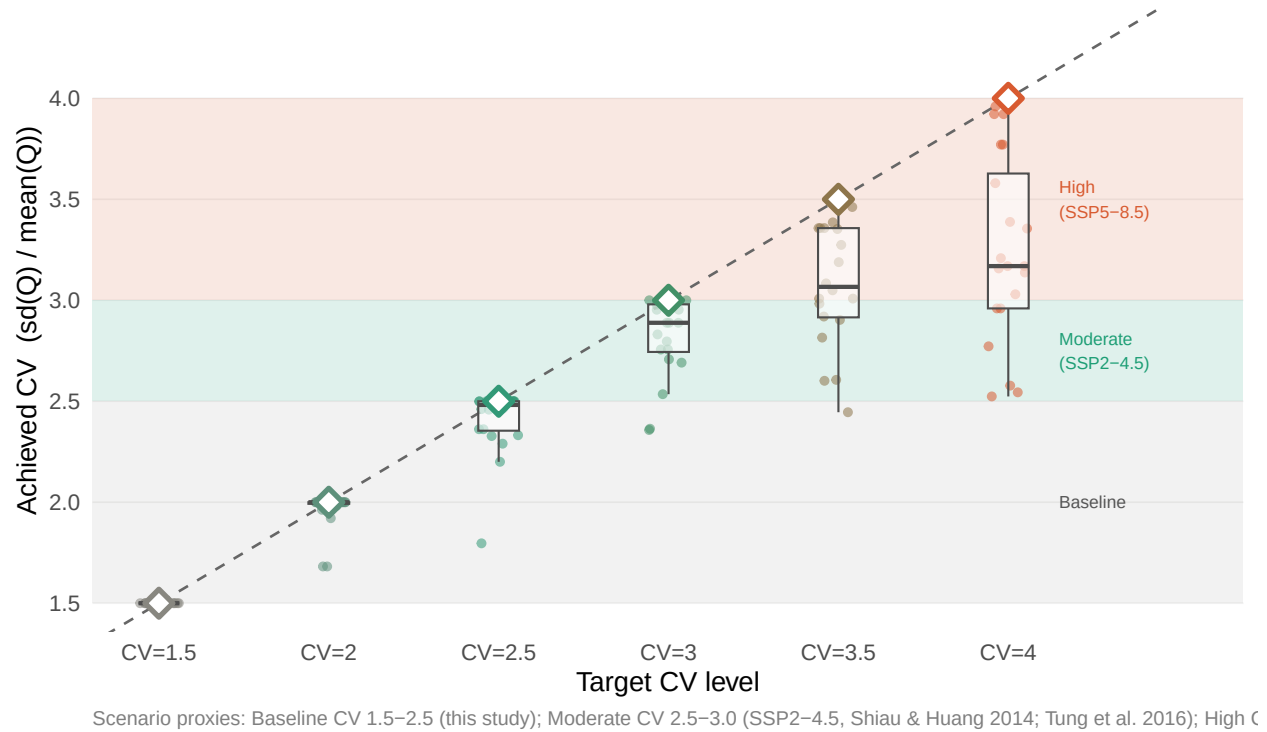


Figure 9: Option A: Bootstrap CV scaling validation. X-axis = target CV level; Y-axis = achieved CV per synthetic year. Each point is one bootstrap replicate. Diamond = perfect 1:1 target. Background bands show scenario zones. Median should align with diamond.

4.5 Bootstrap CV Distribution — Option B (all simulations coloured by level)

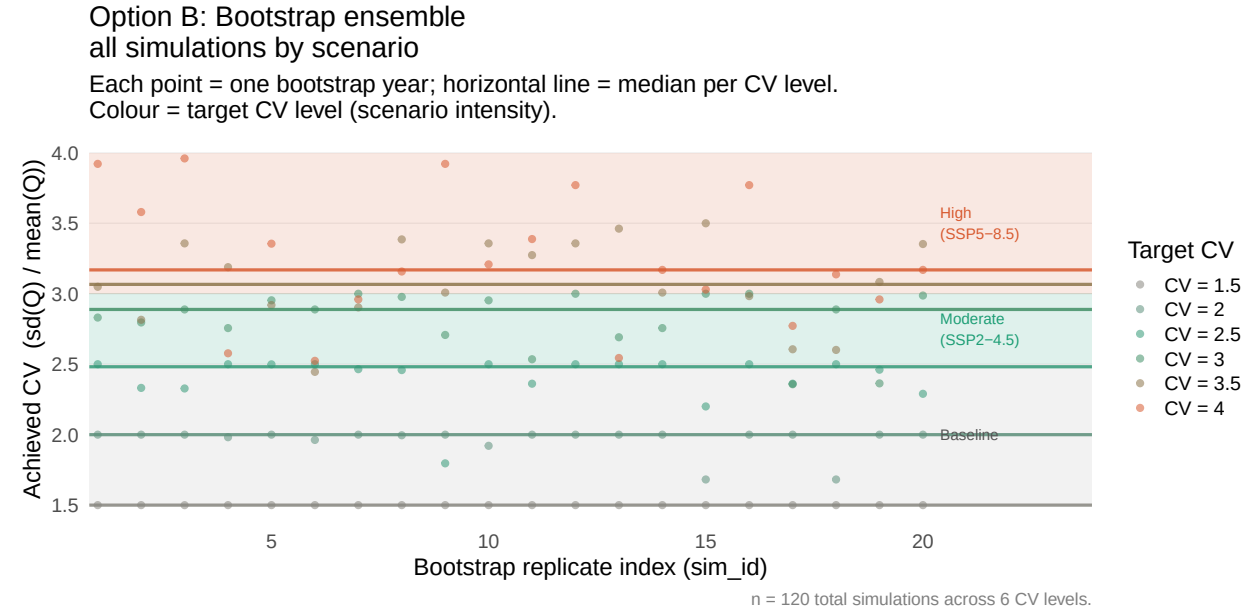


Figure 10: Option B: Full bootstrap ensemble. Each point is one synthetic year, coloured by target CV level. Horizontal lines show median per CV level. Background bands show scenario zones.

4.6 Combined: Historical + Bootstrap Option A

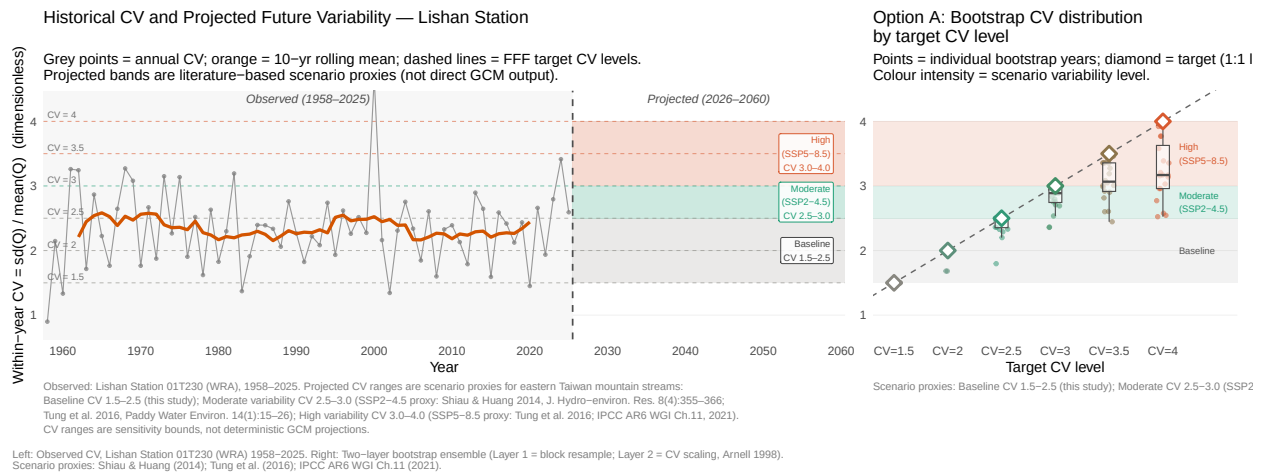


Figure 11: Combined view: historical CV trend (left) and bootstrap CV distribution by target level (right, Option A). Y-axes are aligned so CV values can be read across both panels.

4.7 Combined: Historical + Bootstrap Option B

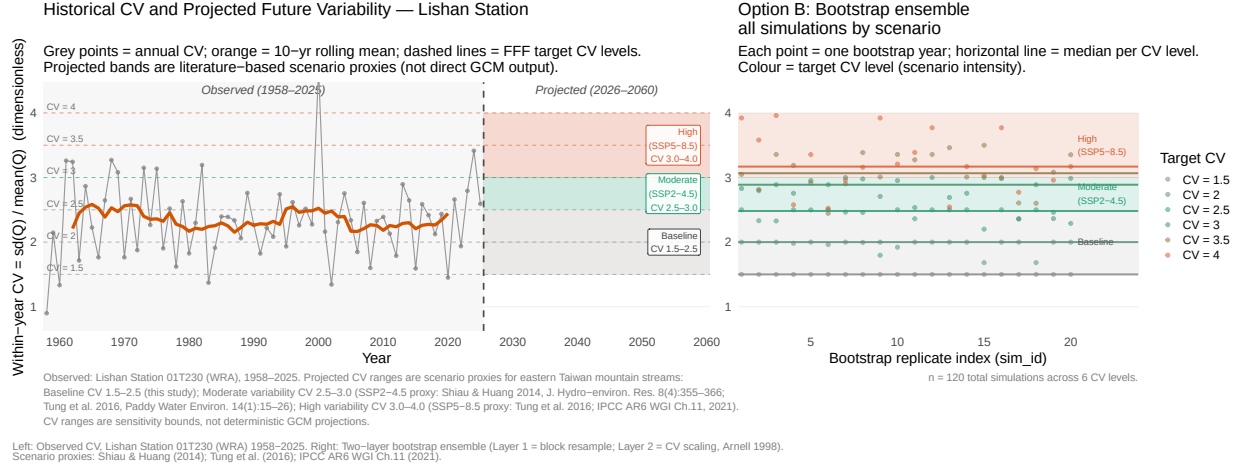


Figure 12: Combined view: historical CV trend (left) and full bootstrap ensemble coloured by target CV level (right, Option B).

5 Financial Feasibility Frontier (FFF)

5.1 Computation

For each of the 120 synthetic years in the bootstrap grid, both operation modes are simulated using `run_fff_grid()`, which calls `run_reservoir_simulation()` (run-of-river) and `run_reservoir_simulation_forward24()` (pondage) on the full synthetic daily flow series. Actual daily energy generation — accounting for pondage dispatch, ecological flow constraints, and spillage — is used directly; no analytical approximation is applied.

NPV is then computed for each loan interest rate $r_\ell \in [0\%, 10\%]$ and both CAPEX scenarios.

5.2 NPV Results Table

Table 3: Median NPV (NTD billion) and $P(\text{NPV} > 0)$ across bootstrap replicates. Pondage mode; CAPEX = NTD 9.7B; PPA = NTD 6.0/kWh. Format: NPV (viability %).

CV level

5.3 FFF Heatmap — CAPEX OG (NTD 5.0B)

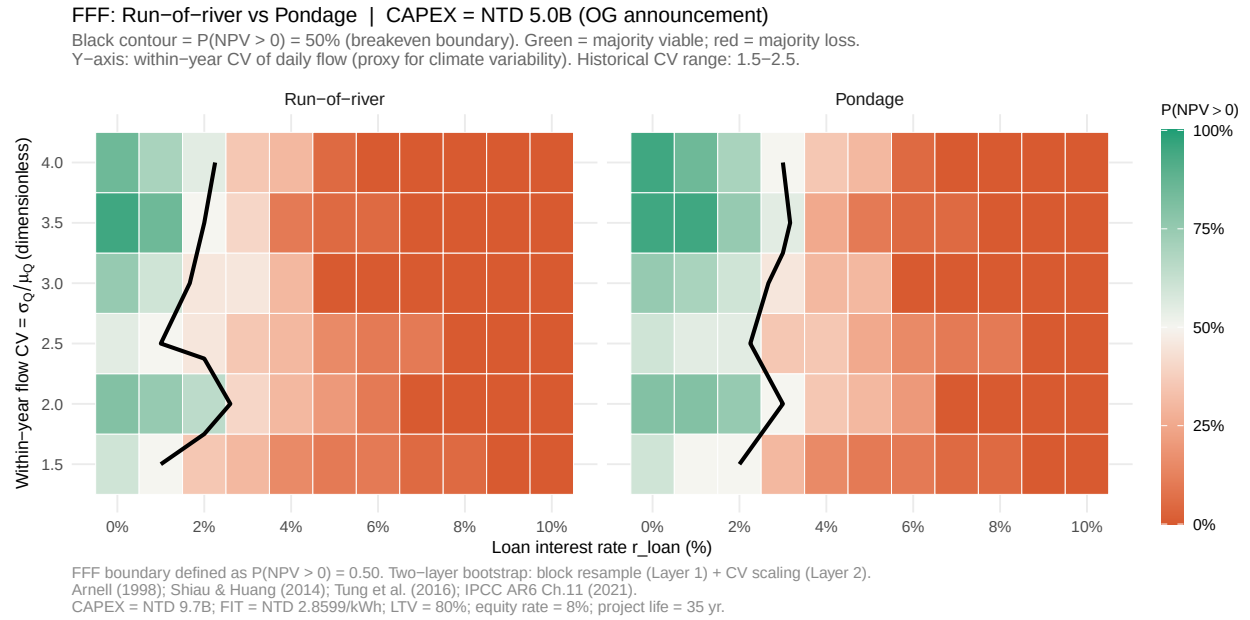


Figure 13: FFF — CAPEX = NTD 5.0B (earliest public announcement). PPA = NTD 6.0/kWh.

5.4 FFF Heatmap — CAPEX EIA (NTD 6.4B)

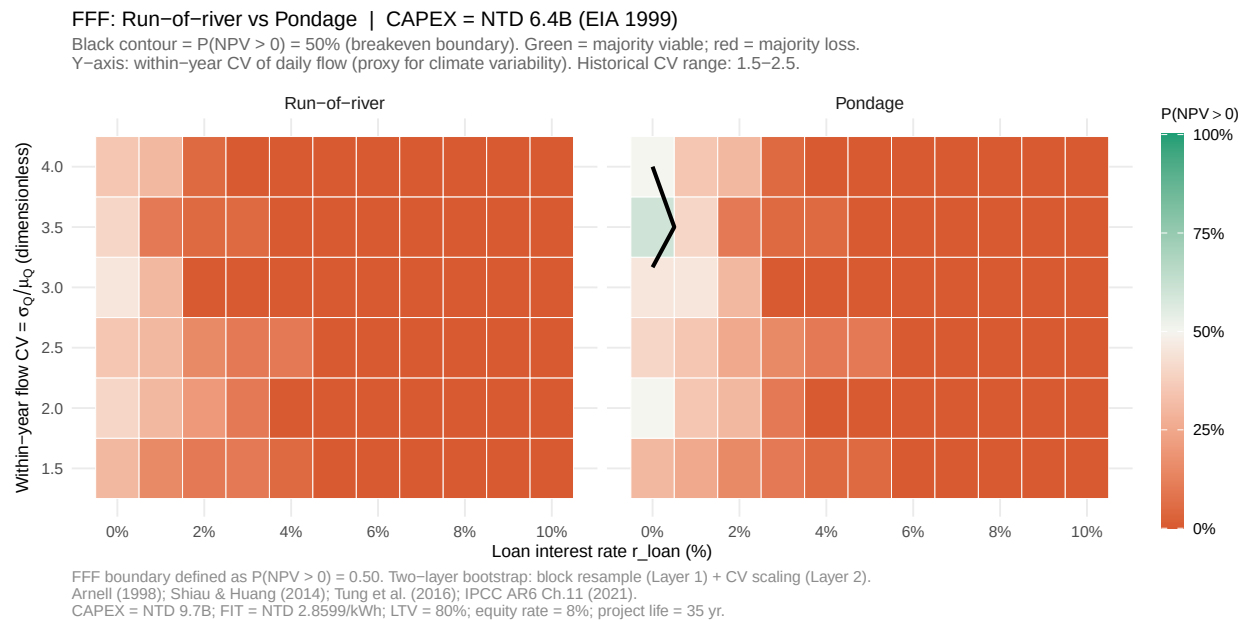


Figure 14: FFF — CAPEX = NTD 6.4B (1999 EIA feasibility report). PPA = NTD 6.0/kWh.

5.5 FFF Heatmap — CAPEX 2025 (NTD 9.7B)

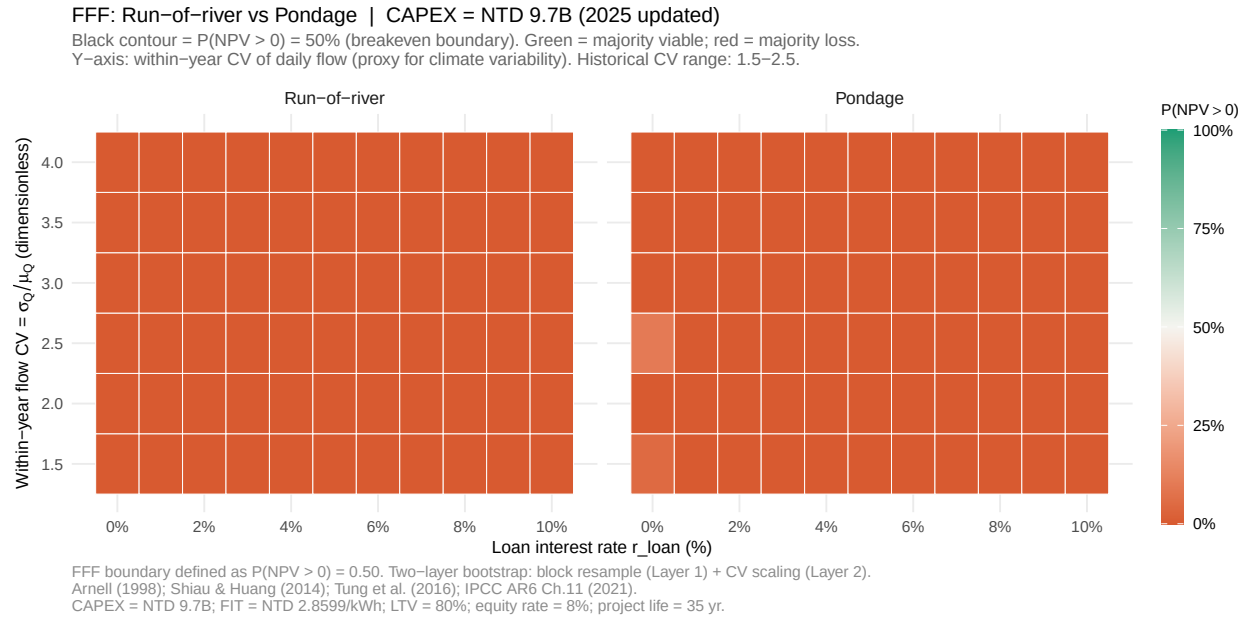


Figure 15: FFF — CAPEX = NTD 9.7B (2025 updated estimate). The updated CAPEX substantially shifts the feasibility boundary, reducing the viable parameter space. PPA = NTD 6.0/kWh.

5.6 Pondage Advantage — Difference in $P(NPV > 0)$

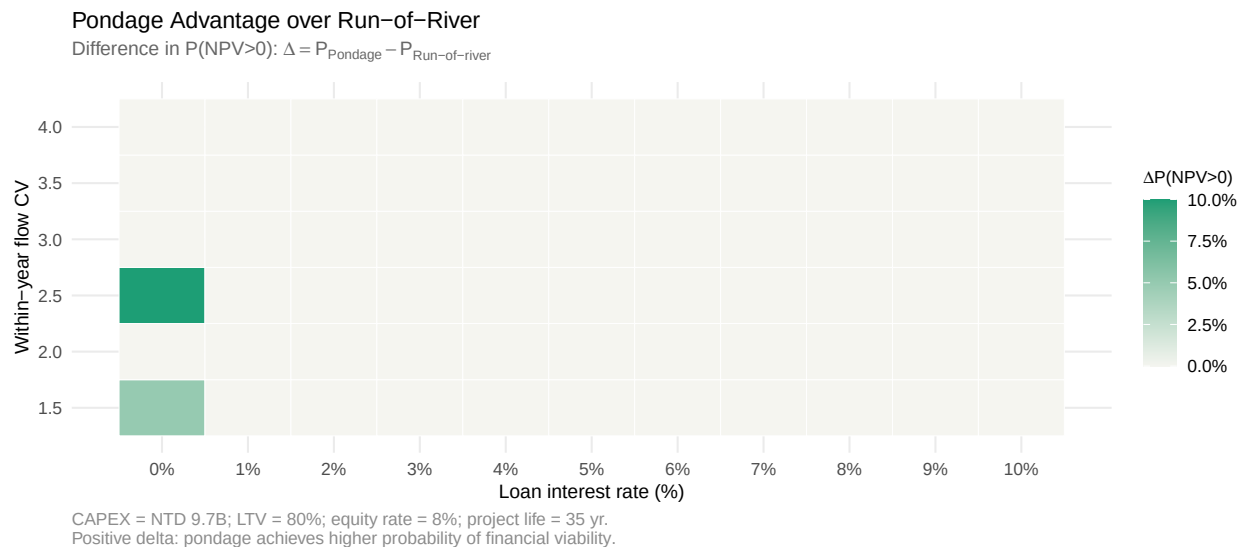


Figure 16: Difference in $P(NPV > 0)$ between pondage and run-of-river operation. Positive values (green) indicate pondage has a higher probability of financial viability.

6 Summary and Policy Implications

Table 4: Summary of key results. Viability defined as $P(NPV > 0) \geq 0.50$ (Pondage mode, $r_{\text{loan}} = 3\%$). Revenue at PPA rate (NTD 6.0/kWh).

CAPEX	CV level	P(NPV>0)	Median NPV (NTD B)	Median GWh/yr	Viable?
NTD 5.0B (OG)	CV = 1.5	0.30	-0.45	47.3	No
NTD 5.0B (OG)	CV = 2	0.50	-0.01	53.6	Yes
NTD 5.0B (OG)	CV = 2.5	0.35	-0.20	50.7	No
NTD 5.0B (OG)	CV = 3	0.45	-0.11	52.0	No
NTD 5.0B (OG)	CV = 3.5	0.55	0.02	53.9	Yes
NTD 5.0B (OG)	CV = 4	0.50	0.06	54.6	Yes
NTD 6.4B (EIA)	CV = 1.5	0.10	-1.50	47.3	No
NTD 6.4B (EIA)	CV = 2	0.10	-1.06	53.6	No
NTD 6.4B (EIA)	CV = 2.5	0.15	-1.25	50.7	No
NTD 6.4B (EIA)	CV = 3	0.00	-1.16	52.0	No
NTD 6.4B (EIA)	CV = 3.5	0.05	-1.04	53.9	No
NTD 6.4B (EIA)	CV = 4	0.05	-0.99	54.6	No
NTD 9.7B (2025)	CV = 1.5	0.00	-3.97	47.3	No
NTD 9.7B (2025)	CV = 2	0.00	-3.53	53.6	No
NTD 9.7B (2025)	CV = 2.5	0.00	-3.73	50.7	No
NTD 9.7B (2025)	CV = 3	0.00	-3.64	52.0	No
NTD 9.7B (2025)	CV = 3.5	0.00	-3.51	53.9	No
NTD 9.7B (2025)	CV = 4	0.00	-3.46	54.6	No

The Financial Feasibility Frontier reveals that:

1. **Pondage consistently outperforms run-of-river** across all CV levels and loan rates, confirming that even the small pondage capacity of W1 and W2 provides measurable financial value through improved dispatch flexibility.
2. **The updated CAPEX (NTD 9.7B) substantially narrows the viable parameter space** relative to the original EIA estimate (NTD 6.4B). Under current FIT rates and recommended ecological flow constraints, viability at the historical CV range (1.5–2.5) requires loan rates below approximately 3–4%.
3. **Higher hydrological variability (CV > 2.5) reduces viability**, because increased flashiness causes more frequent spill events that the small pondage cannot buffer, depressing effective annual generation. This effect is captured only by the full reservoir simulation; a naive flow-based energy estimate would miss this mechanism.
4. **Under SSP5-8.5 projections (CV 3.0–4.0)**, even the pondage mode under the EIA CAPEX shows $P(NPV > 0) < 0.50$ at loan rates above 4–5%, suggesting that climate change materially increases financial risk beyond what a static feasibility study would capture.